

LE_BMS_A_V2

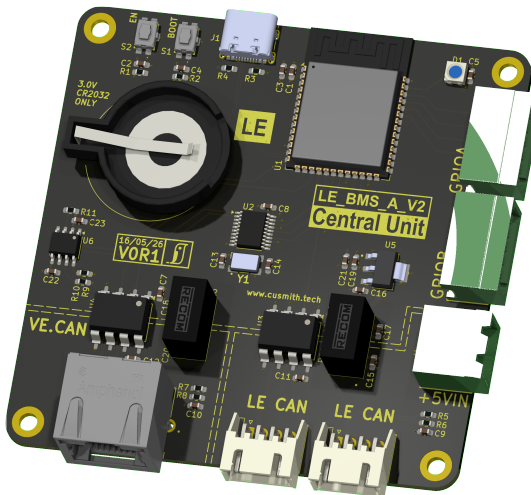
Variant: CHECKED

2026-05-25

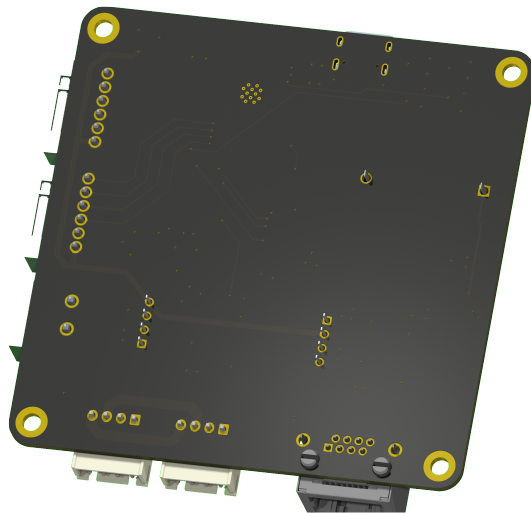
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 25-May-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

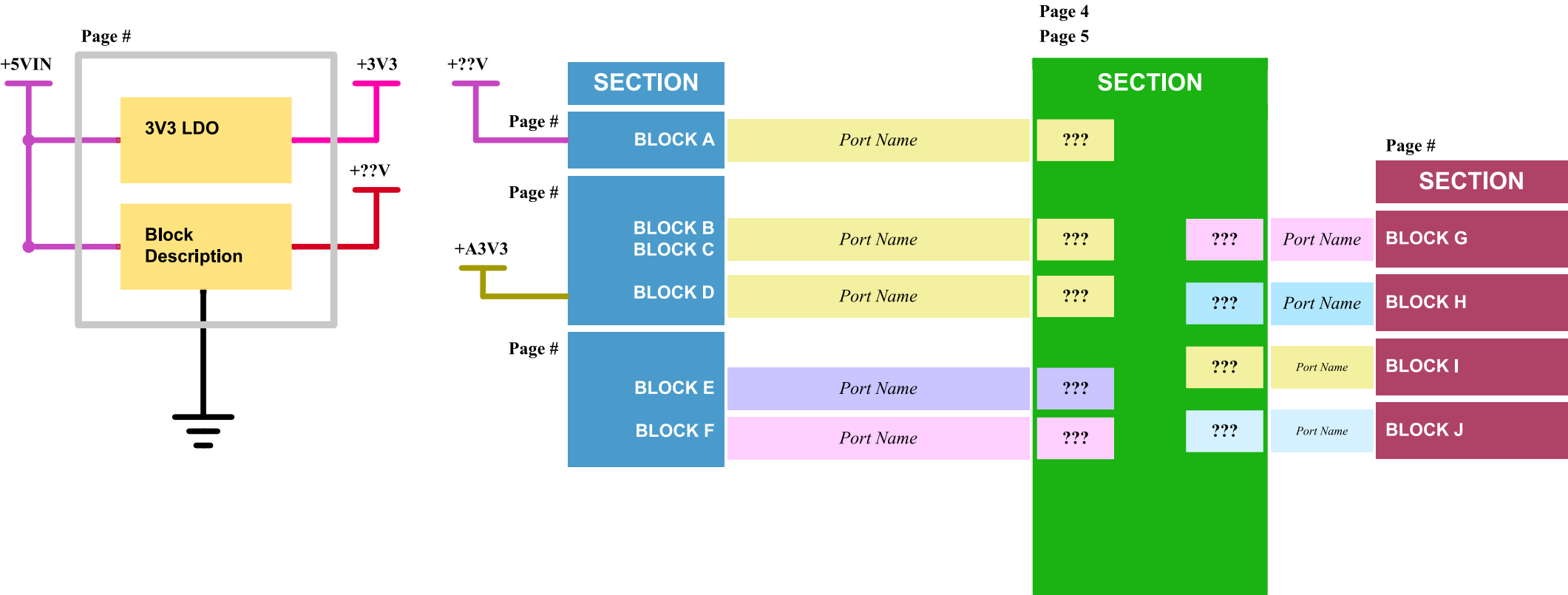
Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

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		Company Name	CHECKED	5f6d398
		Board Name:	Project Name:	
	Sheet Title:	File Name:	Designer:	Date:
		LE_BMS_A_V2.kicad_sch	Author	2025-01-12
	Sheet Path:	Reviewer:	Size:	Revision:
	/		A3	+ (Unreleased)
				1 of 9

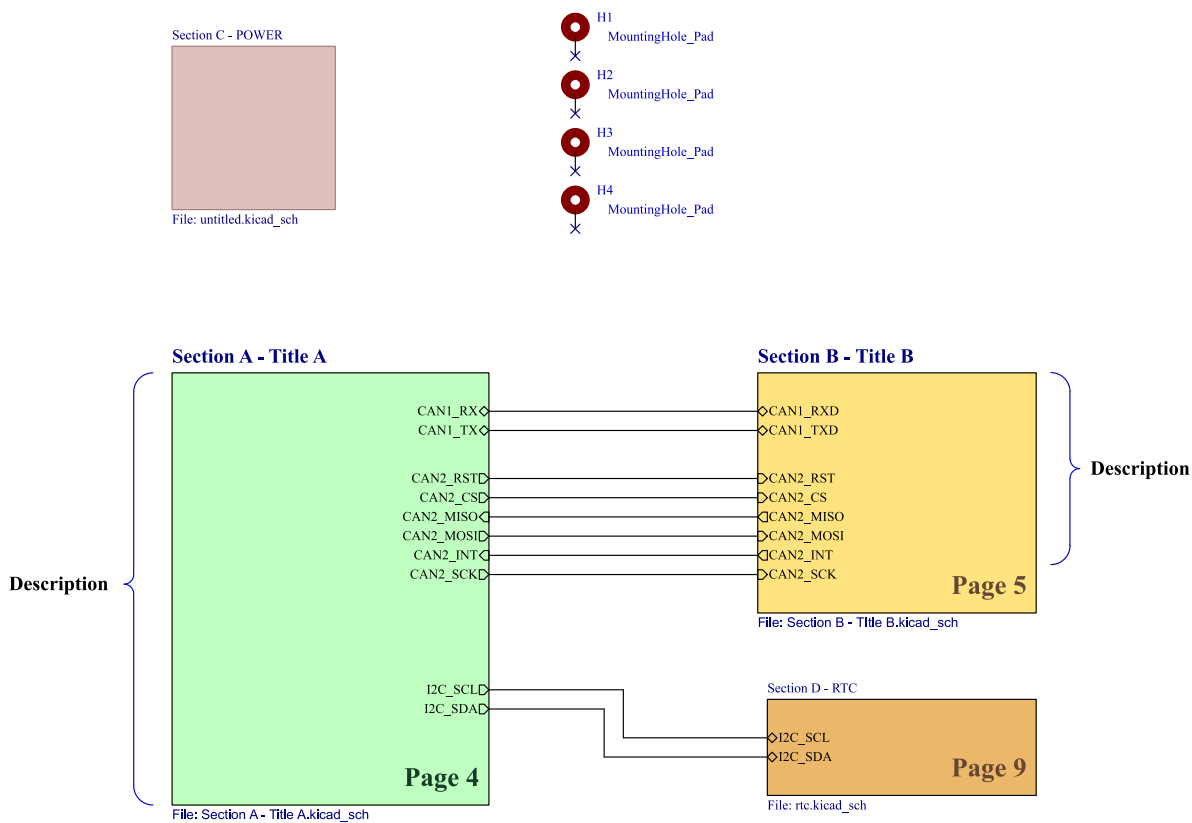
[2] Block Diagram



Target specifications:	
Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

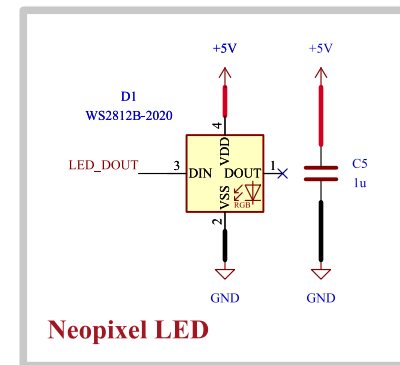
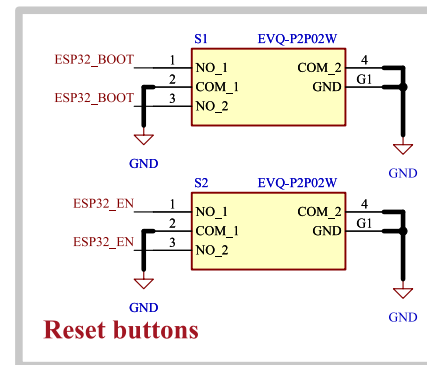
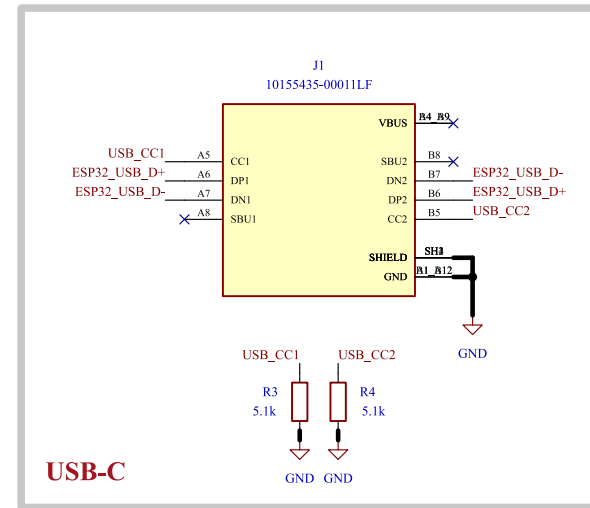
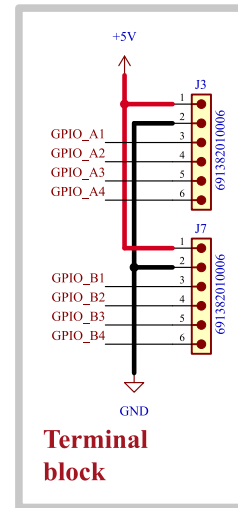
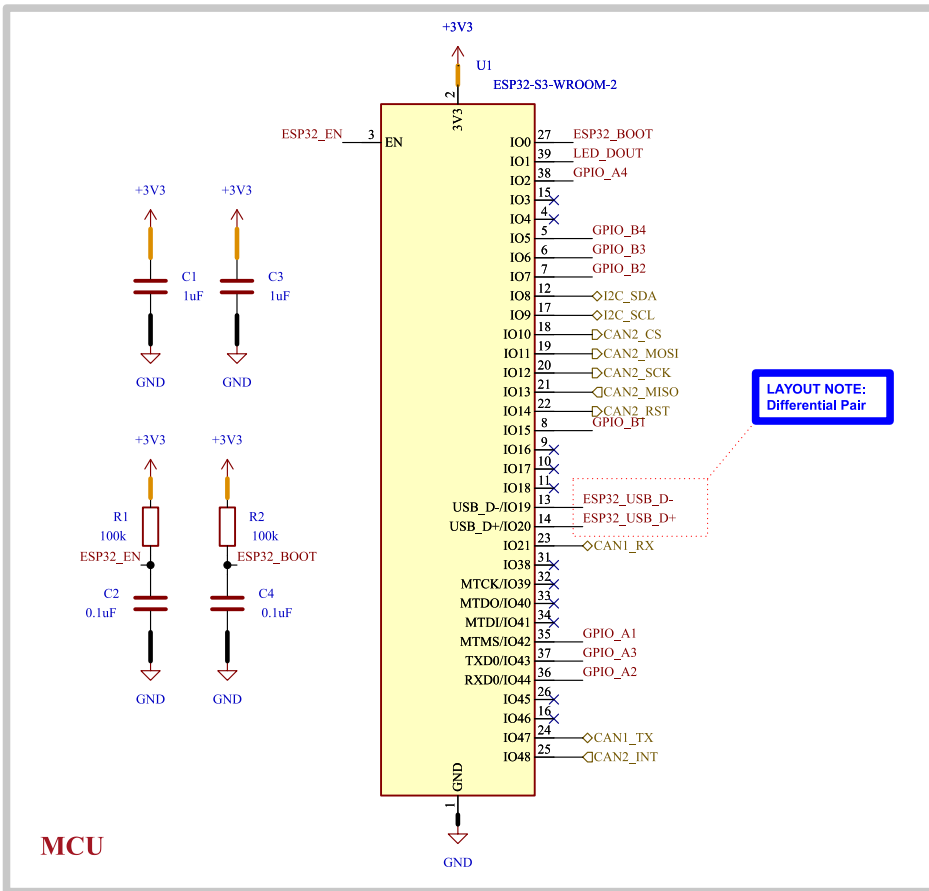
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	Sheet Title: Block Diagram	Board Name: LE_BMS_A_V2		Project Name: LE_V2	
	Sheet Path: /Block Diagram/	File Name: Block Diagram.kicad_sch	Designer: Author	Date: 2025-01-12	Revision: + (Unreleased)
			Reviewer:	Size: A3	Sheet: 2 of 9

[3] Project Architecture



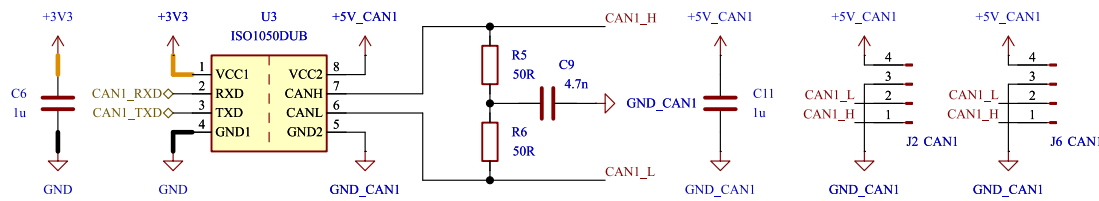
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	Sheet Path: /Project Architecture/	File Name: Project Architecture.kicad_sch	Designer: Author	Date: 2025-01-12	Revision: + (Unreleased)
				Size: A3	Sheet: 3 of 9

[4] MCU, USB and LED

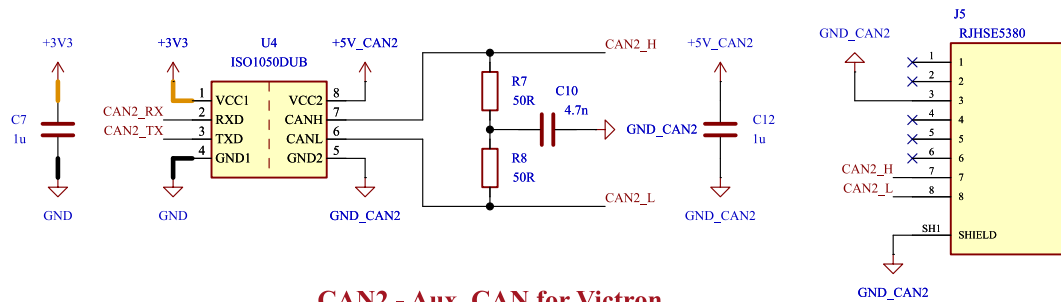


Comments: AN5346 STM32G474 Datasheet p.81 J. Pieper ADC investigation	Company: Company Name		Variant: CHECKED	Git Hash: 5f6d398
	Board Name: LE_BMS_A_V2		Project Name: LE_V2	
	Sheet Title: MCU, USB and LED	File Name: Section A - Title A.kicad_sch	Designer: Author	Date: 2025-01-12
Sheet Path: /Project Architecture/Section A - Title A/		Reviewer:	Size: A4	Revision: + (Unreleased)
			Sheet: 4 of 9	

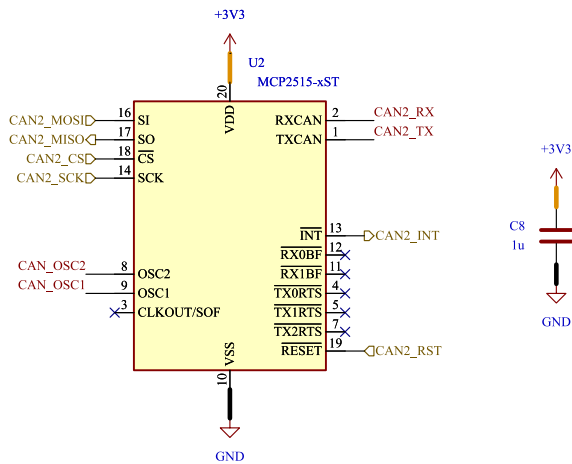
[5] CAN1, CAN2



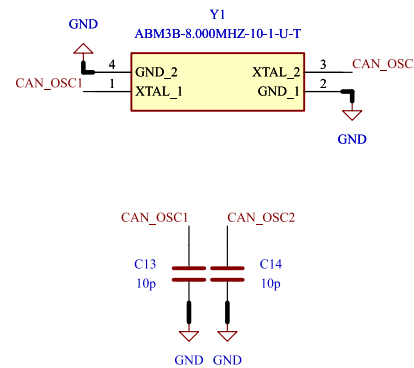
CAN1 - Native ESP32



CAN2 - Aux. CAN for Victron



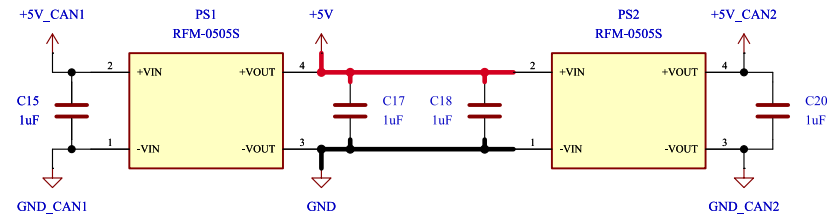
SPI CAN Interface



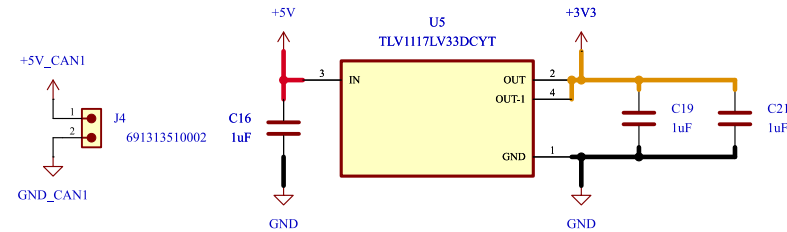
Crystal for CAN

	Comments: References: Flexible I/O worked examples Flexible I/O source configuration		Company: Company Name		Variant: CHECKED	Git Hash: 5f6d398
	Sheet Title: CAN1, CAN2		Board Name: LE_BMS_A_V2		Project Name: LE_V2	
	Sheet Path: /Project Architecture/Section B - Title B/		File Name: Section B - Title B.kicad_sch	Designer: Author	Date: 2025-01-12	Revision: + (Unreleased)
			Reviewer:		Size: A3	Sheet: 5 of 9

[8] Power



5V and 5V_CAN2 Generation



LDO 3V3

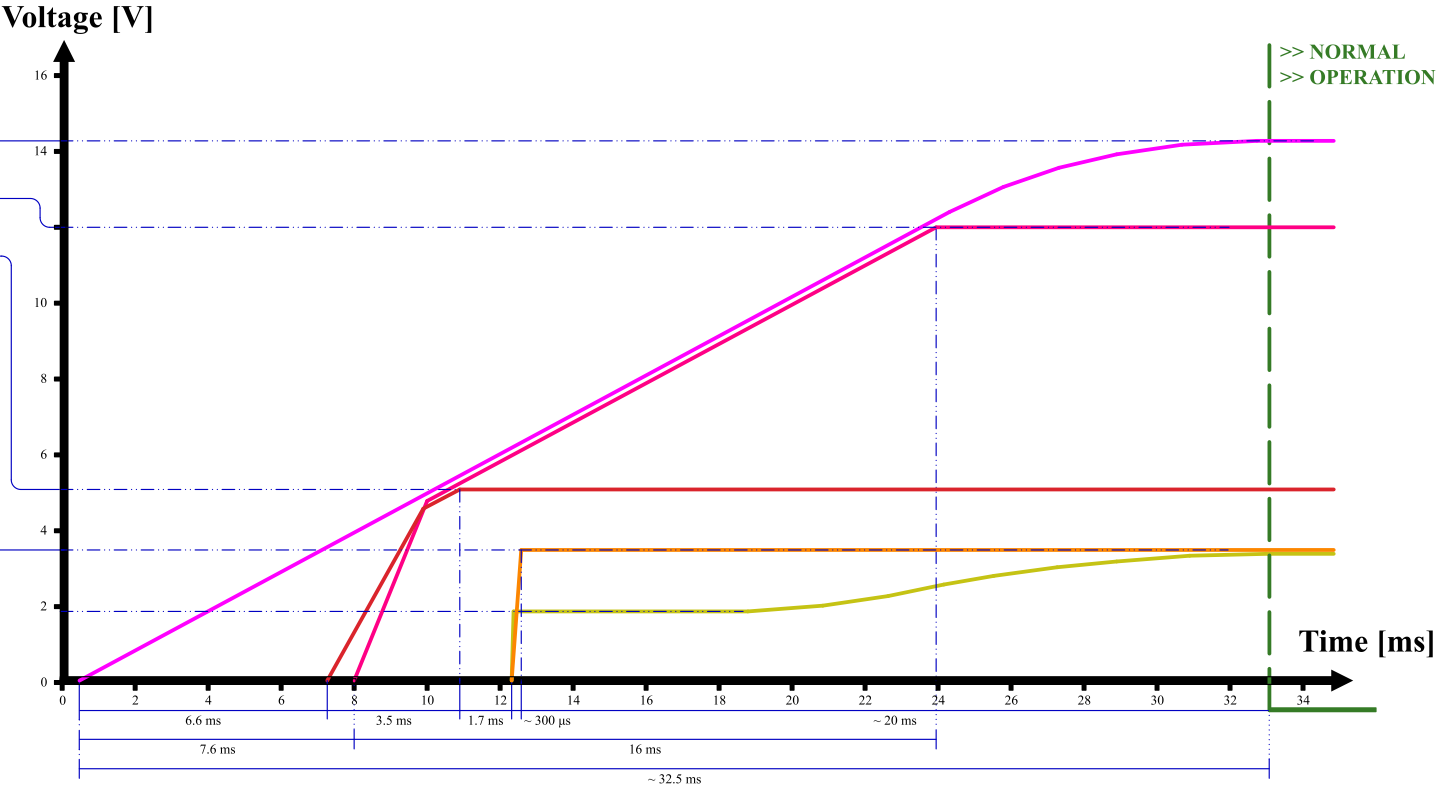
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	Power	LE_BMS_A_V2		LE_V2	
	Sheet Path:	File Name:	Designer:	Date:	Revision:
	/Project Architecture/Section C - POWER/	untitled.kicad_sch	Author		
		Reviewer:	Size:	Sheet:	
			A4	8 of 9	

RTC

	Comments:	Company:		Variant: CHECKED	Git Hash: 5f6d398	
		Board Name: LE_BMS_A_V2		Project Name: LE_V2		
	Sheet Title: RTC	File Name: rtc.kicad_sch	Designer: Author	Date:	Revision:	
	Sheet Path: /Project Architecture/Section D - RTC/		Reviewer:	Size: A4	Sheet: 9 of 9	

[6] Power - Sequencing

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%



	Comments:	Company: Company Name		Variant: CHECKED	Git Hash: 5f6d398
		Board Name: LE_BMS_A_V2		Project Name: LE_V2	
	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Author	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 6 of 9

[7] Revision History

	Comments:	Company: Company Name		Variant: CHECKED	Git Hash: 5f6d398
		Board Name: LE_BMS_A_V2		Project Name: LE_V2	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Author	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 7 of 9